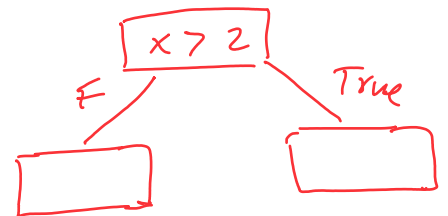
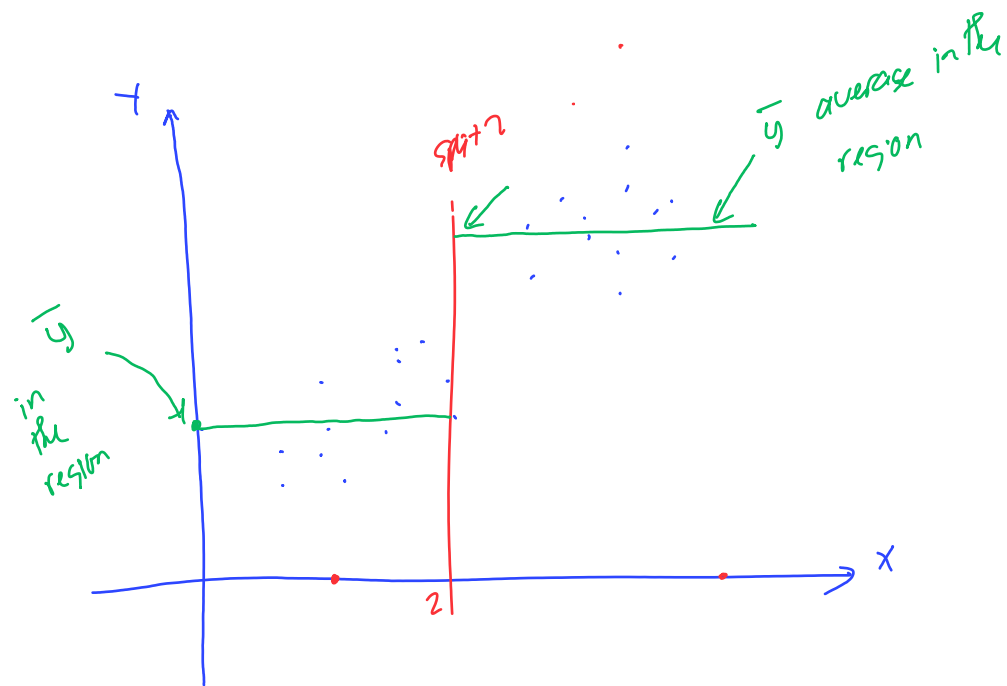
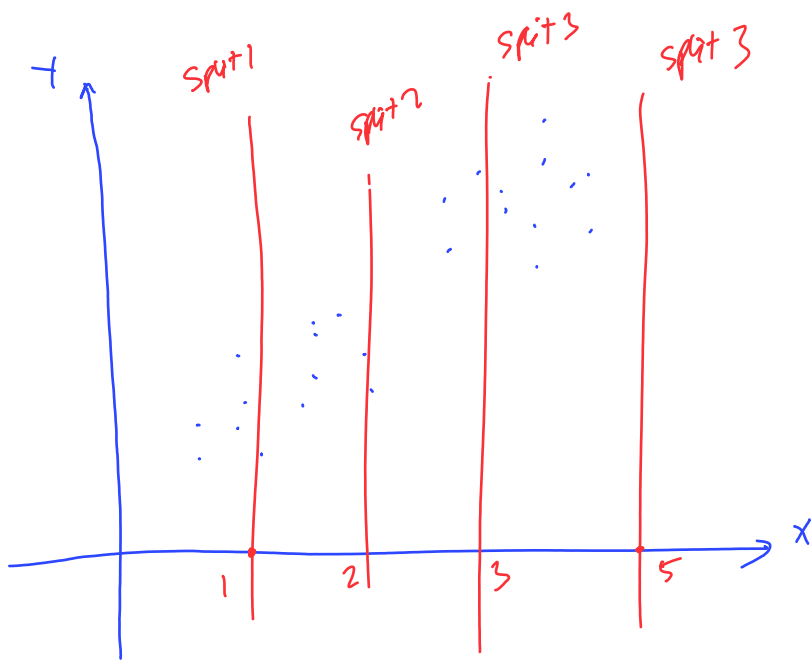
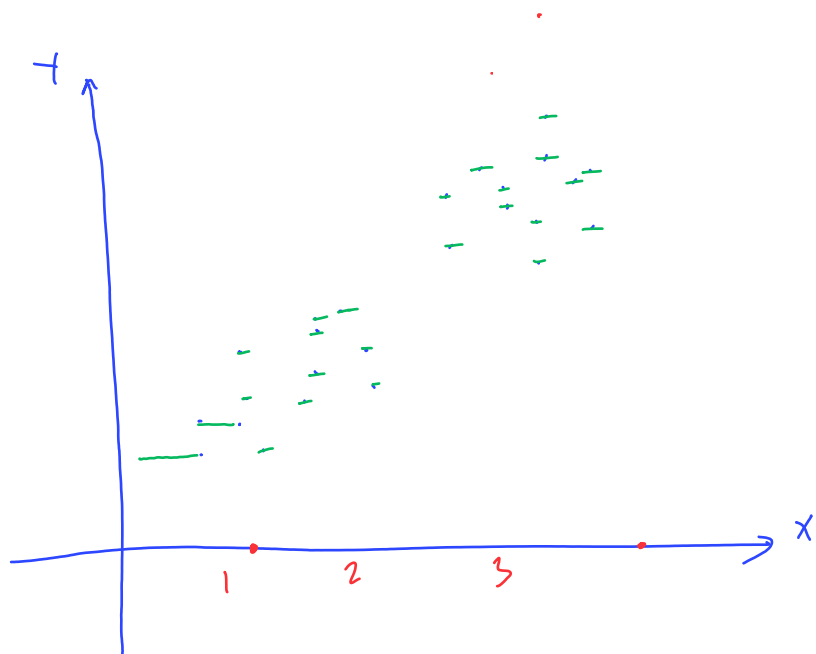
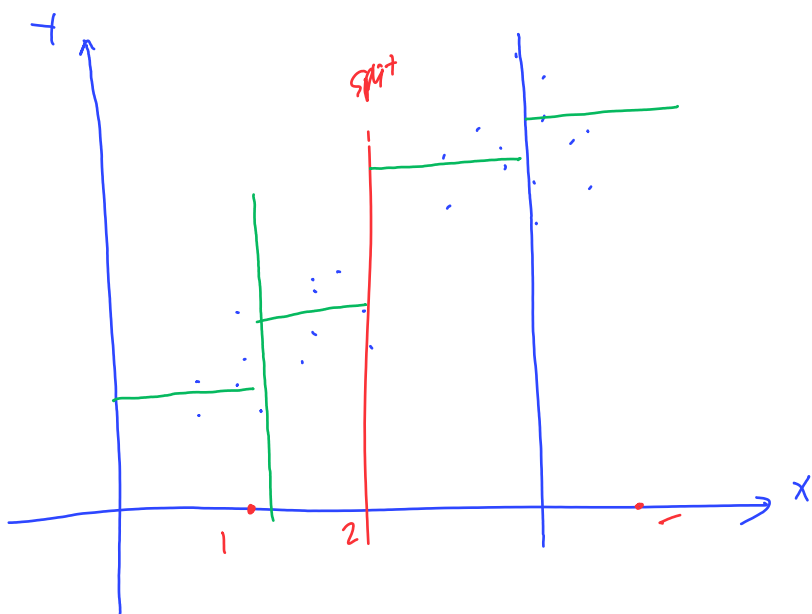


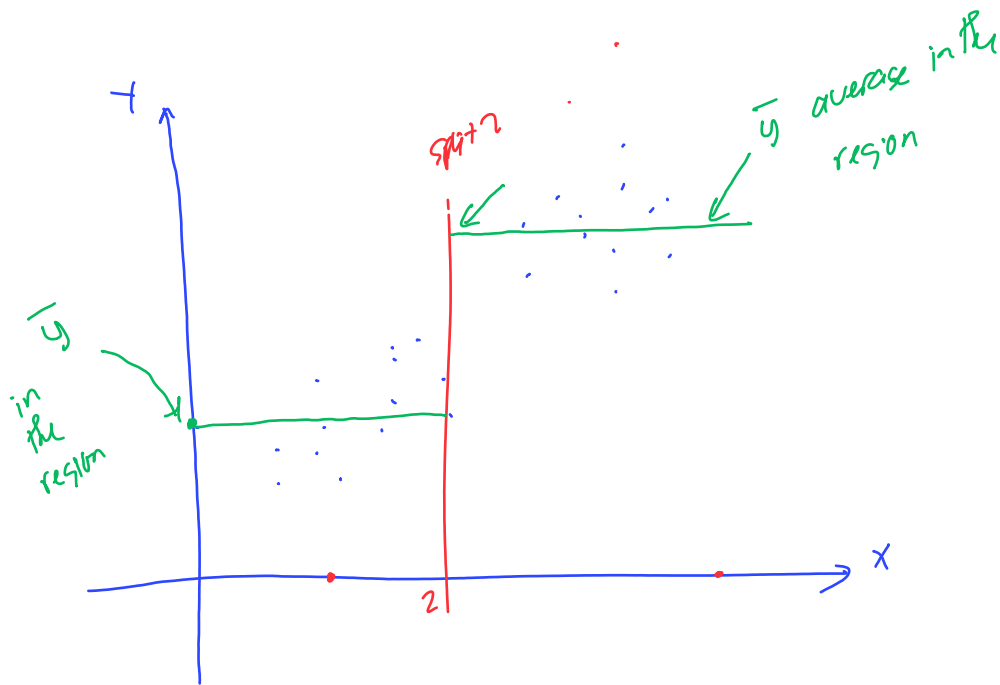
Regression Trees



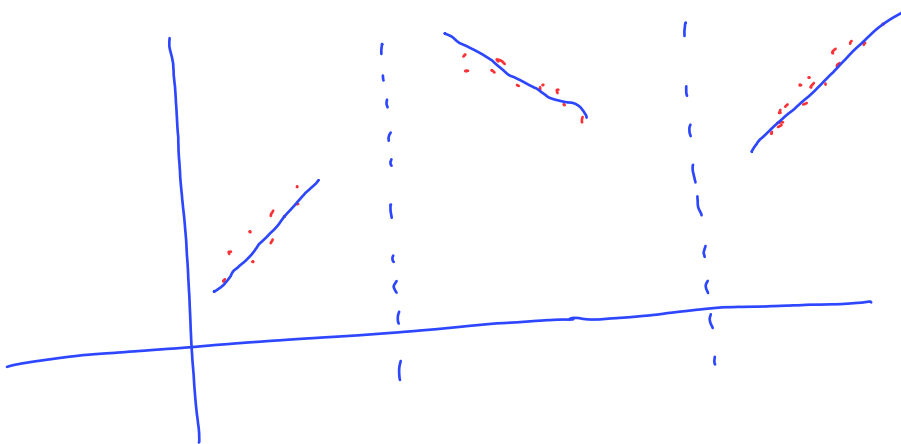
④ prection of the tree is the average values of y in the region.



How do we decide the best split?



we search for the split with the minimum square errors.



Regression Trees

- ▶ The tree will search for all combination of predictors and cutoff value to decide the best split
- ▶ In Regression tree, the best split is the split that minimizes

$$\underbrace{\sum_{i:\mathbf{x}_i \in R_1(j,s)} (y_i - \hat{y}_{R_1})^2}_{\text{RSS of obs. in left branch}} + \underbrace{\sum_{i:\mathbf{x}_i \in R_2(j,s)} (y_i - \hat{y}_{R_2})^2}_{\text{RSS of obs. in right branch}}$$

- ▶ $\hat{y}_{R_1}$ and $\hat{y}_{R_2}$ are the means of the responses falling in to the left branch and right branch, respectively.

Example

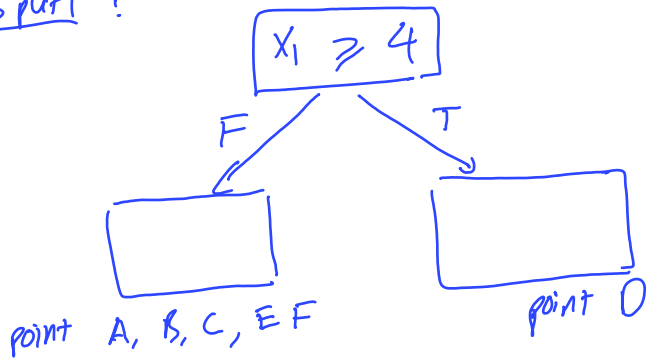
	X_1	X_2	Y
A	1	0	1.2
B	2	1	2.1
C	3	2	1.5
D	4	1	3.0
E	2	2	2.0
F	1	1	1.6

Using the RSS to decide the best split among

- Split 1: Region 1 $X_1 < 4$, Region 2 $X_1 \geq 4$
- Split 2: Region 1 $X_2 < 2$, Region 2 $X_2 \geq 2$

Split 1 :

	X_1	X_2	Y
A	1	0	1.2
B	2	1	2.1
C	3	2	1.5
D	4	1	3.0
E	2	2	2.0
F	1	1	1.6



$$\bar{y}_1 = \frac{y_A + y_B + y_C + y_E + y_F}{5}$$

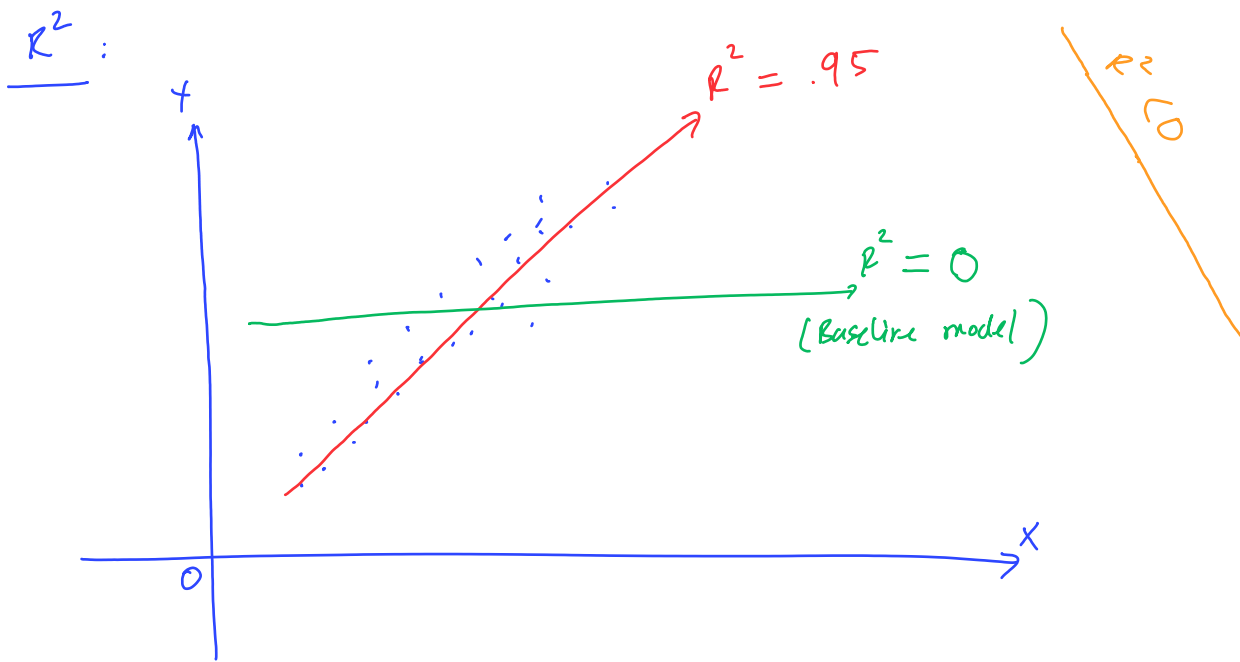
$$\bar{y}_1 = \frac{1.2 + 2.1 + 1.5 + 2.0 + 1.6}{5}$$

$$\bar{y}_1 = 1.68$$

$$\bar{y}_2 = \frac{y_D}{1} = 3.0$$

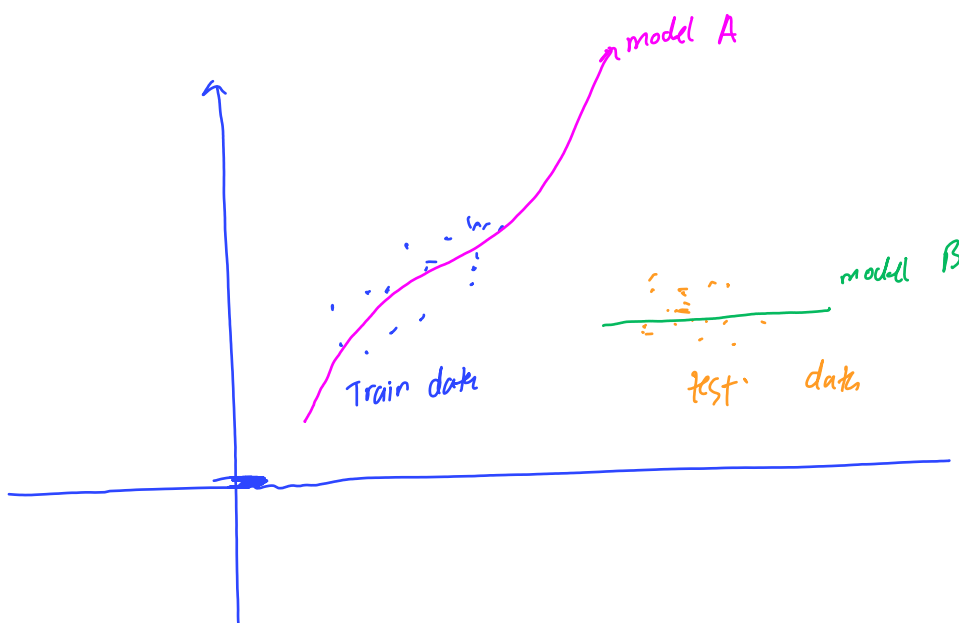
$$RSS = \sum_{\text{region 1}} (y_i - \bar{y}_1)^2 + \sum_{\text{region 2}} (y_i - \bar{y}_2)^2$$

	$\hat{y}$	$(y - \hat{y})^2$
A	1.68	$(1.2 - 1.68)^2$
B	1.68	$(2.1 - 1.68)^2$
C	1.68	$(1.5 - 1.68)^2$
D	3.0	$(3.0 - 3.0)^2$
E	1.68	$(2.0 - 1.68)^2$
F	1.68	$(1.6 - 1.68)^2$
		$\Sigma = RSS$



$$R^2 = 1 - \frac{\text{Sum square error of the model}}{\text{Sum square error of the baseline model}}$$

$$R^2 = 1 - \frac{\sum (y_i - \hat{y}_i)^2}{\sum (y_i - \bar{y})^2}$$



R^2 of model A
on test data
is negative